

POWERBOX · REQUIREMENTS

PowerBox requirements

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DOCUMENT CONTROL

Document ID	PBX-REQ-001	Version	0.4
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/powerbox		
Source	docs/03-requirements.md @ 0a81f90		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	TRL 3. Record Amish's decisions (PBX-DDR-001), including the 230 V AC region and a budget that excludes the pack; SwapCell interface v0.3; status from PBX-CAL-001
0.3	2026-09-24	Amish Chadha	Removed StepGen as a charging source after StepGen became a walking vehicle (Amish, 2026-09-24)
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

PowerBox requirements

These requirements are checked by calculation in PBX-CAL-001 (TRL 3). They are still proposals, not user-validated needs, and will be revised after co-design sessions. On paper, eight are met; **R2 is not met** (1.87 evenings), R6 and R11 are at risk, and R12 cannot be verified until users try a pack swap. The Status column gives the TRL 3 result.

ID	Requirement	Target	Verification (TRL 3 or later)	Status (PBX-CAL-001)
R1	Store useful energy in one SwapCell pack	400 Wh or more usable at the pack terminals, using 10 to 100 % state of charge	Calculation from SwapCell capacity and the host's discharge cut-off	Met, 419 Wh
R2	Run a household through outage evenings	Two evenings of the reference load profile (Table 1, 203 Wh at the loads) on one full pack	Energy budget calculation including conversion losses and standby	Not met , 1.87 evenings
R3	Accept four charging sources	(a) DC input 12 to 60 V, up to 200 W, with maximum power point tracking for a solar panel or another DC source; (b) grid through a certified external charger; (c) SunSpoke by swapping in a charged SwapCell pack; (d) SunSpoke's 100 W panel through the DC input	Design review against the input specification table	Met (design review)
R4	Recharge quickly from the grid	10 to 100 % in 3 h or less	Charger rating and SwapCell charge curve	Met, 2.1 h
R5	Recharge from solar in a day	10 to 100 % in one clear day (4.5 peak sun hours) with a 200 W panel	Solar yield calculation	Met, full in 0.70 day
R6	Use DC and solar input efficiently at low power	DC input conversion efficiency 90 % or more from 40 to 200 W, and stable maximum power point tracking (no stall or hunting) as a panel's output changes with light and temperature	Converter efficiency data; later bench test with a solar panel or solar simulator	At risk, 90.5 % worst case in the loss model
R7	Provide the outputs households use	2 x USB-C PD (one 100 W, one 60 W), 2 x USB-A (12 W each), 2 x 12 V DC (10 A shared), 1 x AC outlet, 230 V 50 Hz, 300 W continuous pure sine (THD 3 % or less) behind a 30 mA RCD; total output limited to 400 W	Design review against the output specification table	Met (design review)
R8	Waste little energy when idle	Off: 5 % or less of pack energy per month, including cell self-discharge. Ready (display on, outputs off): 1.0 W or less. Inverter switches itself off after 10 min below 5 W	Standby budget calculation; later measurement	Met, 3.2 % per month off, 0.6 W ready
R9	Never back-feed household wiring	No AC input connector that can mate with an AC output; AC output only through a 30 mA RCD outlet with the inverter neutral bonded to the case; grid charging only through a certified external charger with a DC plug; DC bus 60 V or less	Design review and safety checklist	Met (design review)
R10	Be carried by one adult	10 kg (22 lb) or less with pack; 500 x 300 x 280 mm or smaller including handle; one top handle	Massing model, then weighing	Met, 8.6 kg, 482 x 276 x 278 mm
R11	Stay within the concept budget	PowerBox parts \$450 or less, including the grid charger and excluding the SwapCell pack, which is	Priced BOM	At risk, \$449

ID	Requirement	Target	Verification (TRL 3 or later)	Status (PBX-CAL-001)
		priced once in the SwapCell BOM (decided by Amish, 2026-09-25)		
R12	Swap packs and show status simply	Pack swapped by hand in 30 s or less without tools; display shows state of charge, input and output power and estimated time remaining, read from the SwapCell interface v0.3 CAN messages; a sleeping pack wakes through the INTERLOCK loop (v0.3 item W)	Design review; later timed trial with users	Not verifiable at TRL 3

■ Assumptions

Table 1. Reference evening load profile (6 pm to 11 pm outage). Proposed for review; the real profile must come from co-design sessions.

Load	Power	Hours	Energy at the load	Output used
Three LED bulbs	3 x 5 W	5	75 Wh	12 V DC
Wi-Fi router and fibre terminal	12 W	5	60 Wh	12 V DC
Radio	5 W	4	20 Wh	12 V DC
Four phone charges	about 12 Wh each		48 Wh	USB
Total			203 Wh	

- The SwapCell pack is the reference 13S2P pack of SwapCell interface v0.3: 46.8 V nominal, 39.0 to 54.6 V, 468 Wh nameplate and 466 Wh at 0.2C (SWC-CAL-001). PowerBox is a station host (v0.3 item C) and its receptacle carries the 10 k Ω INTERLOCK coding resistor (item W).
- Solar input spends much of the day well below the panel rating (morning, evening and cloud), so the DC input often runs at 40 to 100 W.
- Refrigerators, kettles and other high-power or high-surge loads are out of scope (see PBX-PRB-001).
- Budget: the \$450 in `project.yaml` covers the PowerBox itself (decided by Amish, 2026-09-25, PBX-DDR-001). A household that needs its own pack also buys a SwapCell pack, priced once in the SwapCell BOM at about \$414 in prototype parts.
- R2: whether to relax R2, lower the cut-off to 3.5 % or revise the profile is proposed, awaiting Amish (PBX-DDR-001 item 13).